Local Interpretability Report

Customer ID: 100002

Default Probability (TARGET=1): 0.7269

Decision explanation

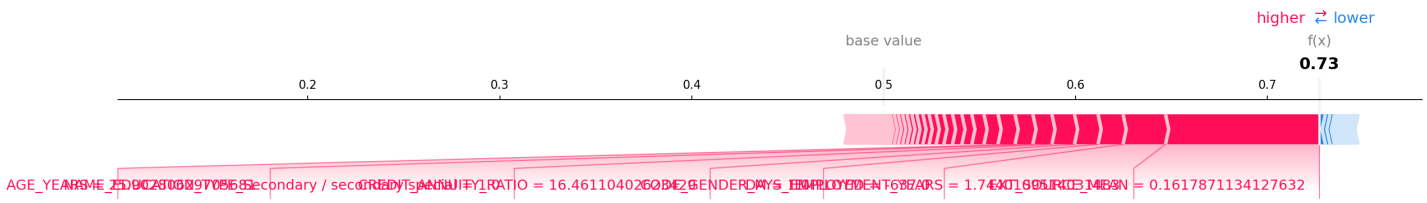
The model predicts default with default probability 72.69%. The class decision uses a 0.50 probability threshold (ties go to class 0). The SHAP baseline default probability is 50.00%; positive contributions increase default risk and negative contributions reduce it. Strongest increases in risk: EXT_SOURCE_MEAN (+8.01 percentage points); EMPLOYMENT_YEARS (+2.21 percentage points); DAYS_EMPLOYED (+1.30 percentage points). Strongest reductions in risk: EMERGENCYSTATE_MODE_No (-0.32 percentage points); DAYS_LAST_PHONE_CHANGE (-0.24 percentage points); CREDIT_INCOME_RATIO (-0.23 percentage points). This agrees with the observed label 1 (default).

Largest SHAP contributions to default probability

Feature	Client value	Contribution (percentage points)
EXT_SOURCE_MEAN	0.1618	+8.01
EMPLOYMENT_YEARS	1.7440	+2.21
DAYS_EMPLOYED	-637.0000	+1.30
CODE_GENDER_M	1.0000	+1.23
CREDIT_ANNUITY_RATIO	16.4611	+1.18
NAME_EDUCATION_TYPE_Secondary / secondary special	1.0000	+0.99
AGE_YEARS	25.9028	+0.93
AMT_GOODS_PRICE	351000.0000	+0.89

Interactive Plotly client and population charts: client1_correct_train.html

SHAP Force Plot



Client Data vs. Population Average

Feature	Client Value	Dataset Average
CNT_CHILDREN	0.00	0.42
AMT_GOODS_PRICE	351000.00	538316.29
REGION_POPULATION_RELATIVE	0.02	0.02
DAYS_EMPLOYED	-637.00	-2251.61
DAYS_REGISTRATION	-3648.00	-4986.12
DAYS_ID_PUBLISH	-2120.00	-2994.20
FLAG_MOBIL	1.00	1.00
FLAG_EMP_PHONE	1.00	0.82
FLAG_WORK_PHONE	0.00	0.20
FLAG_CONT_MOBILE	1.00	1.00
FLAG_PHONE	1.00	0.28
FLAG_EMAIL	0.00	0.06
CNT_FAM_MEMBERS	1.00	2.15
REGION_RATING_CLIENT	2.00	2.05
REGION_RATING_CLIENT_W_CITY	2.00	2.03
HOUR_APPR_PROCESS_START	10.00	12.06
REG_REGION_NOT_LIVE_REGION	0.00	0.02
REG_REGION_NOT_WORK_REGION	0.00	0.05
LIVE_REGION_NOT_WORK_REGION	0.00	0.04
REG_CITY_NOT_LIVE_CITY	0.00	0.08

REG_CITY_NOT_WORK_CITY	0.00	0.23
LIVE_CITY_NOT_WORK_CITY	0.00	0.18
APARTMENTS_AVG	0.02	0.10
BASEMENTAREA_AVG	0.04	0.08
YEARS_BEGINEXPLUATATION_AVG	0.97	0.98
ELEVATORS_AVG	0.00	0.04
ENTRANCES_AVG	0.07	0.14
FLOORSMAX_AVG	0.08	0.20
LANDAREA_AVG	0.04	0.06
LIVINGAREA_AVG	0.02	0.09
NONLIVINGAREA_AVG	0.00	0.01
APARTMENTS_MODE	0.03	0.10
BASEMENTAREA_MODE	0.04	0.08
YEARS_BEGINEXPLUATATION_MODE	0.97	0.98
ELEVATORS_MODE	0.00	0.03
ENTRANCES_MODE	0.07	0.14
FLOORSMAX_MODE	0.08	0.19
LANDAREA_MODE	0.04	0.05
LIVINGAREA_MODE	0.02	0.09
NONLIVINGAREA_MODE	0.00	0.01
APARTMENTS_MEDI	0.03	0.10
BASEMENTAREA_MEDI	0.04	0.08

YEARS_BEGINEXPLUATATION_MEDI	0.97	0.98
ELEVATORS_MEDI	0.00	0.04
ENTRANCES_MEDI	0.07	0.14
FLOORSMAX_MEDI	0.08	0.20
LANDAREA_MEDI	0.04	0.06
LIVINGAREA_MEDI	0.02	0.09
NONLIVINGAREA_MEDI	0.00	0.01
TOTALAREA_MODE	0.01	0.09
OBS_30_CNT_SOCIAL_CIRCLE	2.00	1.42
DEF_30_CNT_SOCIAL_CIRCLE	2.00	0.14
OBS_60_CNT_SOCIAL_CIRCLE	2.00	1.40
DEF_60_CNT_SOCIAL_CIRCLE	2.00	0.10
DAYS_LAST_PHONE_CHANGE	-1134.00	-962.86
FLAG_DOCUMENT_2	0.00	0.00
FLAG_DOCUMENT_3	1.00	0.71
FLAG_DOCUMENT_4	0.00	0.00
FLAG_DOCUMENT_5	0.00	0.02
FLAG_DOCUMENT_6	0.00	0.09
FLAG_DOCUMENT_7	0.00	0.00
FLAG_DOCUMENT_8	0.00	0.08
FLAG_DOCUMENT_9	0.00	0.00
FLAG_DOCUMENT_10	0.00	0.00

FLAG_DOCUMENT_11	0.00	0.00
FLAG_DOCUMENT_12	0.00	0.00
FLAG_DOCUMENT_13	0.00	0.00
FLAG_DOCUMENT_14	0.00	0.00
FLAG_DOCUMENT_15	0.00	0.00
FLAG_DOCUMENT_16	0.00	0.01
FLAG_DOCUMENT_17	0.00	0.00
FLAG_DOCUMENT_18	0.00	0.01
FLAG_DOCUMENT_19	0.00	0.00
FLAG_DOCUMENT_20	0.00	0.00
FLAG_DOCUMENT_21	0.00	0.00
AMT_REQ_CREDIT_BUREAU_HOUR	0.00	0.01
AMT_REQ_CREDIT_BUREAU_DAY	0.00	0.01
AMT_REQ_CREDIT_BUREAU_WEEK	0.00	0.03
AMT_REQ_CREDIT_BUREAU_MON	0.00	0.23
AMT_REQ_CREDIT_BUREAU_QRT	0.00	0.23
AMT_REQ_CREDIT_BUREAU_YEAR	1.00	1.78
AGE_YEARS	25.90	43.91
EMPLOYMENT_YEARS	1.74	6.16
CREDIT_INCOME_RATIO	2.01	3.96
ANNUITY_INCOME_RATIO	0.12	0.18
CREDIT_ANNUITY_RATIO	16.46	21.61

INCOME_PER_PERSON	202500.00	93105.76
EXT_SOURCE_MEAN	0.16	0.51
NAME_CONTRACT_TYPE_Cash loans	1.00	0.90
NAME_CONTRACT_TYPE_Revolving loans	0.00	0.10
CODE_GENDER_F	0.00	0.66
CODE_GENDER_M	1.00	0.34
CODE_GENDER_XNA	0.00	0.00
FLAG_OWN_CAR_N	1.00	0.66
FLAG_OWN_CAR_Y	0.00	0.34
FLAG_OWN_REALTY_N	0.00	0.31
FLAG_OWN_REALTY_Y	1.00	0.69
NAME_TYPE_SUITE_Children	0.00	0.01
NAME_TYPE_SUITE_Family	0.00	0.13
NAME_TYPE_SUITE_Group of people	0.00	0.00
NAME_TYPE_SUITE_Missing	0.00	0.00
NAME_TYPE_SUITE_Other_A	0.00	0.00
NAME_TYPE_SUITE_Other_B	0.00	0.01
NAME_TYPE_SUITE_Spouse, partner	0.00	0.04
NAME_TYPE_SUITE_Unaccompanied	1.00	0.81
NAME_INCOME_TYPE_Businessman	0.00	0.00

NAME_INCOME_TYPE_Commercial associate	0.00	0.23
NAME_INCOME_TYPE_Maternity leave	0.00	0.00
NAME_INCOME_TYPE_Pensioner	0.00	0.18
NAME_INCOME_TYPE_State servant	0.00	0.07
NAME_INCOME_TYPE_Student	0.00	0.00
NAME_INCOME_TYPE_Unemployed	0.00	0.00
NAME_INCOME_TYPE_Working	1.00	0.52
NAME_EDUCATION_TYPE_Academic degree	0.00	0.00
NAME_EDUCATION_TYPE_Higher education	0.00	0.24
NAME_EDUCATION_TYPE_Incomplete higher	0.00	0.03
NAME_EDUCATION_TYPE_Lower secondary	0.00	0.01
NAME_EDUCATION_TYPE_Secondary / secondary special	1.00	0.71
NAME_FAMILY_STATUS_Civil marriage	0.00	0.10
NAME_FAMILY_STATUS_Married	0.00	0.64
NAME_FAMILY_STATUS_Separated	0.00	0.06
NAME_FAMILY_STATUS_Single / not married	1.00	0.15
NAME_FAMILY_STATUS_Unknown	0.00	0.00
NAME_FAMILY_STATUS_Widow	0.00	0.05

NAME_HOUSING_TYPE_Co-op apartment	0.00	0.00
NAME_HOUSING_TYPE_House / apartment	1.00	0.89
NAME_HOUSING_TYPE_Municipal apartment	0.00	0.04
NAME_HOUSING_TYPE_Office apartment	0.00	0.01
NAME_HOUSING_TYPE_Rented apartment	0.00	0.02
NAME_HOUSING_TYPE_With parents	0.00	0.05
OCCUPATION_TYPE_Accountants	0.00	0.03
OCCUPATION_TYPE_Cleaning staff	0.00	0.02
OCCUPATION_TYPE_Cooking staff	0.00	0.02
OCCUPATION_TYPE_Core staff	0.00	0.09
OCCUPATION_TYPE_Drivers	0.00	0.06
OCCUPATION_TYPE_HR staff	0.00	0.00
OCCUPATION_TYPE_High skill tech staff	0.00	0.04
OCCUPATION_TYPE_IT staff	0.00	0.00
OCCUPATION_TYPE_Laborers	1.00	0.18
OCCUPATION_TYPE_Low-skill Laborers	0.00	0.01
OCCUPATION_TYPE_Managers	0.00	0.07
OCCUPATION_TYPE_Medicine staff	0.00	0.03
OCCUPATION_TYPE_Missing	0.00	0.31

OCCUPATION_TYPE_Private service staff	0.00	0.01
OCCUPATION_TYPE_Realty agents	0.00	0.00
OCCUPATION_TYPE_Sales staff	0.00	0.10
OCCUPATION_TYPE_Secretaries	0.00	0.00
OCCUPATION_TYPE_Security staff	0.00	0.02
OCCUPATION_TYPE_Waiters/barmen staff	0.00	0.00
WEEKDAY_APPR_PROCESS_START_FRIDAY	0.00	0.16
WEEKDAY_APPR_PROCESS_START_MONDAY	0.00	0.16
WEEKDAY_APPR_PROCESS_START_SATURDAY	0.00	0.11
WEEKDAY_APPR_PROCESS_START_SUNDAY	0.00	0.05
WEEKDAY_APPR_PROCESS_START_THURSDAY	0.00	0.16
WEEKDAY_APPR_PROCESS_START_TUESDAY	0.00	0.18
WEEKDAY_APPR_PROCESS_START_WEDNESDAY	0.00	0.17
ORGANIZATION_TYPE_Advertising	0.00	0.00
ORGANIZATION_TYPE_Agriculture	0.00	0.01
ORGANIZATION_TYPE_Bank	0.00	0.01
ORGANIZATION_TYPE_Business Entity Type 1	0.00	0.02
ORGANIZATION_TYPE_Business Entity Type 2	0.00	0.03
ORGANIZATION_TYPE_Business Entity Type 3	1.00	0.22
ORGANIZATION_TYPE_Cleaning	0.00	0.00

ORGANIZATION_TYPE_Construction	0.00	0.02
ORGANIZATION_TYPE_Culture	0.00	0.00
ORGANIZATION_TYPE_Electricity	0.00	0.00
ORGANIZATION_TYPE_Emergency	0.00	0.00
ORGANIZATION_TYPE_Government	0.00	0.03
ORGANIZATION_TYPE_Hotel	0.00	0.00
ORGANIZATION_TYPE_Housing	0.00	0.01
ORGANIZATION_TYPE_Industry: type 1	0.00	0.00
ORGANIZATION_TYPE_Industry: type 10	0.00	0.00
ORGANIZATION_TYPE_Industry: type 11	0.00	0.01
ORGANIZATION_TYPE_Industry: type 12	0.00	0.00
ORGANIZATION_TYPE_Industry: type 13	0.00	0.00
ORGANIZATION_TYPE_Industry: type 2	0.00	0.00
ORGANIZATION_TYPE_Industry: type 3	0.00	0.01
ORGANIZATION_TYPE_Industry: type 4	0.00	0.00
ORGANIZATION_TYPE_Industry: type 5	0.00	0.00
ORGANIZATION_TYPE_Industry: type 6	0.00	0.00
ORGANIZATION_TYPE_Industry: type 7	0.00	0.00

ORGANIZATION_TYPE_Industry: type 8	0.00	0.00
ORGANIZATION_TYPE_Industry: type 9	0.00	0.01
ORGANIZATION_TYPE_Insurance	0.00	0.00
ORGANIZATION_TYPE_Kindergarten	0.00	0.02
ORGANIZATION_TYPE_Legal Services	0.00	0.00
ORGANIZATION_TYPE_Medicine	0.00	0.04
ORGANIZATION_TYPE_Military	0.00	0.01
ORGANIZATION_TYPE_Mobile	0.00	0.00
ORGANIZATION_TYPE_Other	0.00	0.05
ORGANIZATION_TYPE_Police	0.00	0.01
ORGANIZATION_TYPE_Postal	0.00	0.01
ORGANIZATION_TYPE_Realtor	0.00	0.00
ORGANIZATION_TYPE_Religion	0.00	0.00
ORGANIZATION_TYPE_Restaurant	0.00	0.01
ORGANIZATION_TYPE_School	0.00	0.03
ORGANIZATION_TYPE_Security	0.00	0.01
ORGANIZATION_TYPE_Security Ministries	0.00	0.01
ORGANIZATION_TYPE_Self-employed	0.00	0.12
ORGANIZATION_TYPE_Services	0.00	0.01
ORGANIZATION_TYPE_Telecom	0.00	0.00

ORGANIZATION_TYPE_Trade: type 1	0.00	0.00
ORGANIZATION_TYPE_Trade: type 2	0.00	0.01
ORGANIZATION_TYPE_Trade: type 3	0.00	0.01
ORGANIZATION_TYPE_Trade: type 4	0.00	0.00
ORGANIZATION_TYPE_Trade: type 5	0.00	0.00
ORGANIZATION_TYPE_Trade: type 6	0.00	0.00
ORGANIZATION_TYPE_Trade: type 7	0.00	0.03
ORGANIZATION_TYPE_Transport: type 1	0.00	0.00
ORGANIZATION_TYPE_Transport: type 2	0.00	0.01
ORGANIZATION_TYPE_Transport: type 3	0.00	0.00
ORGANIZATION_TYPE_Transport: type 4	0.00	0.02
ORGANIZATION_TYPE_University	0.00	0.00
ORGANIZATION_TYPE_XNA	0.00	0.18
HOUSETYPE_MODE_Missing	0.00	0.50
HOUSETYPE_MODE_block of flats	1.00	0.49
HOUSETYPE_MODE_specific housing	0.00	0.00
HOUSETYPE_MODE_terraced house	0.00	0.00
WALLSMATERIAL_MODE_Block	0.00	0.03

WALLSMATERIAL_MODE_Missing	0.00	0.51
WALLSMATERIAL_MODE_Mixed	0.00	0.01
WALLSMATERIAL_MODE_Monolithic	0.00	0.01
WALLSMATERIAL_MODE_Others	0.00	0.01
WALLSMATERIAL_MODE_Panel	0.00	0.21
WALLSMATERIAL_MODE_Stone, brick	1.00	0.21
WALLSMATERIAL_MODE_Wooden	0.00	0.02
EMERGENCYSTATE_MODE_Missing	0.00	0.47
EMERGENCYSTATE_MODE_No	1.00	0.52
EMERGENCYSTATE_MODE_Yes	0.00	0.01